

L11c: Markov Decision Processes

Value iteration and random rollout

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Objectives for Today

How should an agent choose its actions when each action changes both the reward it collects and the state it moves to?

Today's concepts

- Formulate a Markov decision process: define its states, actions, rewards, transition model, and discount factor, and the policy that chooses an action in each state.
- Solve for the optimal policy with value iteration: update the value of each state from the states it can reach next until the values stop changing.
- Estimate values with random rollouts: average the discounted rewards of simulated random trajectories, without writing down the transition probabilities.

Lecture: Markov Decision Processes (MDPs)

Lecture and Example

Lecture: Markov Decision Processes (MDPs)
In this lecture, we formulate Markov decision processes, solve them with value iteration, and estimate values with random rollouts.

Example: Value Iteration and Random Rollout in the Slippery Grid World
In this example, we solve the lecture's grid world with value iteration, estimate its values with random rollouts, and compare the two answers.

A rollout needs only a simulator; value iteration needs the full transition model.

From Markov Chains to Decisions

A Markov decision process (MDP) extends the Markov chain by adding actions and rewards.

- A Markov chain evolves passively, according to fixed transition probabilities.
- In an MDP, an agent chooses actions that influence the state transitions and generate rewards.

MDPs are the foundation for sequential decision-making under uncertainty: the agent must make good decisions, one after another, in a stochastic environment.

MDP Components

An MDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, R(s,a), T(s' \mid s,a), \gamma)$:

- State space $\mathcal{S}$: all possible states s of the system.
- Action space $\mathcal{A}$: all possible actions a; the actions available in state s are $\mathcal{A}_s \subseteq \mathcal{A}$.
- Reward function $R(s,a)$: the expected immediate reward of action a in state s. One transition produces an observed reward r.
- Transition model $T(s' \mid s,a) = P(s_{t+1} = s' \mid s_t = s, a_t = a)$: the probability that action a in state s leads to state s'.
- Discount factor $\gamma \in [0,1)$: weights future rewards against immediate ones; $\gamma = 0$ cares only about now, and $\gamma \to 1$ values long-term returns.

Policies and Terminal States

The agent's goal is a policy $\pi : \mathcal{S} \to \mathcal{A}$, with $\pi(s) = a$, that maximizes the expected cumulative discounted reward:

$$\mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)\right].$$

Previously, π denoted a state distribution; here it is a policy.

In an episodic problem, entering a terminal state ends reward collection:

- The reward for reaching the terminal state is collected on entry.
- Value iteration treats it as absorbing: a dummy action keeps the agent there and earns zero reward, so its value stays zero.

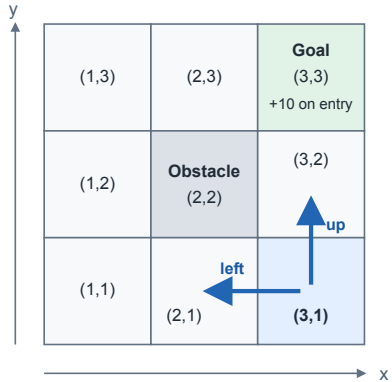
Today: value iteration solves for the optimal policy, and random rollout estimates the value of a state by simulation.

Example: Grid World Navigation

A robot navigates a 3×3 grid to reach a goal while avoiding an obstacle and minimizing travel time:

- States: the eight squares (x, y), $x, y \in \{1, 2, 3\}$, other than the obstacle at $(2, 2)$; x increases to the right and y upward.
- Actions: up, down, left, right; moves off the grid are unavailable, so $\mathcal{A}_s \subsetneq \mathcal{A}$ on the edges.
- Rewards: $+10$ for entering the goal at $(3, 3)$, -5 for a bump into the obstacle, and -1 for every other outcome, including a slip off the grid.
- Transitions: slippery terrain; the intended move succeeds with probability 0.8, and each perpendicular move happens with probability 0.1.
- Discount factor $\gamma = 0.9$: the $+10$ goal outweighs the -1 step costs, so the robot pursues the goal.

Grid World: The Layout



Update the corner (3,1)

Two available actions: up and left

Each action has three outcomes

0.8: move in the intended direction

0.1: slip in each perpendicular direction

Rewards

Goal entry: +10; no further reward

Obstacle bump: −5; remain in place

Other outcomes: −1

A slip off the grid also leaves the robot in place.

Because moves are random, $R(s, a)$ averages over where the robot actually moves. A bump or a slip off the grid leaves it in place, and the goal is terminal.

Value Iteration

Value iteration is a dynamic programming algorithm: it builds the value of each state from the values of the states that can follow it. It repeats the Bellman backup:

$$U_{k+1}(s) = \max_{a \in \mathcal{A}_s} \left(R(s,a) + \gamma \sum_{s' \in \mathcal{S}} T(s' \mid s, a)\, U_k(s') \right)$$

- For $0 \leq \gamma < 1$, $U_k(s) \to U^*(s)$ as $k \to \infty$.
- The optimal value function $U^*(s)$ is the maximum expected cumulative discounted reward achievable from state s under the best possible policy.

Worked Example: One Bellman Backup

Update the corner $s = (3, 1)$, whose available actions are up and left. Every outcome of either action earns -1.

Starting from $U_0 = 0$, the first sweep gives:

$$U_1(3, 2) = 7.4, \qquad U_1(2, 1) = -1.4, \qquad U_1(3, 1) = -1.$$

For example, 7.4 comes from moving up from $(3, 2)$:

$$0.8\,(+10) + 0.1\,(-5) + 0.1\,(-1) = 7.4,$$

reaching the goal with probability 0.8, bumping the obstacle with probability 0.1, and slipping off the grid with probability 0.1.

Worked Example: Comparing the Two Actions

$Q_1(s, a)$ is the expected return of taking a and then using the values U_1. Up reaches $(3, 2)$ with probability 0.8, slips to $(2, 1)$ or stays with probability 0.1 each; left swaps the two neighbors:

$$Q_1((3,1), \text{up}) = -1 + 0.9(0.8(7.4) + 0.1(-1.4) + 0.1(-1)) = 4.112,$$
$$Q_1((3,1), \text{left}) = -1 + 0.9(0.8(-1.4) + 0.1(7.4) + 0.1(-1)) = -1.432.$$

The Bellman backup takes the larger:

$$\boxed{U_2(3,1) = \max\{4.112,\, -1.432\} = 4.112}$$

The greedy action is up. Every state in the next sweep uses the previous sweep's values in the same way.

Value Iteration Algorithm

Initialize: Given an MDP, a tolerance $\epsilon > 0$, and a maximum number of iterations $k_{\max}$, set $k \leftarrow 0$, $U_0(s) \leftarrow 0$ for all $s \in \mathcal{S}$, and `converged` $\leftarrow$ `false`.

While not `converged` **do**:

1. For each state $s \in \mathcal{S}$, compute the Bellman backup $U_{k+1}(s)$.
2. Check whether to stop:
   - If $\max_{s \in \mathcal{S}} |U_{k+1}(s) - U_k(s)| \leq \epsilon$ **then** set `converged` $\leftarrow$ `true` and return $U^* \leftarrow U_{k+1}$.
   - Otherwise, if $k + 1 \geq k_{\max}$ **then** return U_{k+1} with `converged` $=$ `false` and a warning.
3. Increment the iteration counter, $k \leftarrow k + 1$.

Extract Policy: for each state $s \in \mathcal{S}$,

$$\pi^*(s) \leftarrow \arg\max_{a \in \mathcal{A}_s} \left(R(s,a) + \gamma \sum_{s' \in \mathcal{S}} T(s' \mid s, a)\, U^*(s') \right).$$

Value Iteration: Cost

Each iteration costs $\mathcal{O}\left(|\mathcal{S}|^2\,|\mathcal{A}|\right)$ operations:

- for each of the $|\mathcal{S}|$ states, we evaluate $|\mathcal{A}|$ actions,
- and each action sums over $|\mathcal{S}|$ possible next states.

The number of iterations depends on γ and ϵ: each iteration shrinks the largest error in the value function by at least a factor of γ.

Value iteration is tractable for thousands of states, but challenging for continuous or very large discrete state spaces. There, we need function approximation or sampling-based methods.

Random Rollout

Random rollout estimates the value of a state by simulating random trajectories from it and averaging their cumulative discounted rewards.

- Value iteration needs the transition model $T(s' \mid s, a)$ and computes the values of all states at once.
- Random rollout is model-free: it needs only an environment that can simulate an action and report the next state and reward.

Averaging the returns of many rollouts that start from s gives an empirical estimate $\hat{U}(s)$ of its value.

Random Rollout Algorithm

Initialize: Given a simulated environment, $\gamma \in [0, 1)$, a maximum depth $d \geq 1$, and a number of rollouts $N \geq 1$, set $\hat{U}(s) \leftarrow 0$ and $n(s) \leftarrow 0$ for all $s \in \mathcal{S}$.

For $i = 1$ to N **do**:

1. Choose a starting state s_0; set $s \leftarrow s_0$, $t \leftarrow 0$, $G \leftarrow 0$, and `terminated` $\leftarrow$ `true` if s_0 is terminal, otherwise `false`.
2. While not `terminated` **do**:
   - Select a uniformly at random from $\mathcal{A}_s$; execute it and observe r and s'.
   - Update $G \leftarrow G + \gamma^t r$ and $t \leftarrow t + 1$.
   - If $t \geq d$ or s' is terminal **then** set `terminated` $\leftarrow$ `true`; otherwise, $s \leftarrow s'$.
3. Update the estimate with an incremental mean:

$$n(s_0) \leftarrow n(s_0) + 1, \qquad \hat{U}(s_0) \leftarrow \hat{U}(s_0) + \frac{1}{n(s_0)} \left(G - \hat{U}(s_0) \right).$$

Output: $\hat{U}(s)$ for every state that served as a starting state.

What Does a Rollout Estimate?

As the number of rollouts from s_0 grows, the average return converges to the expected discounted return of the random policy, which picks each available action with equal probability, stopped after d steps or at a terminal state.

This is not the optimal value $U^*(s_0)$ that value iteration computes.

For bounded rewards $|r| \le r_{\max}$, the depth limit omits at most

$$\left|U^{\text{random}}(s) - U_d^{\text{random}}(s)\right| \le \sum_{t=d}^{\infty} \gamma^t r_{\max} = \frac{\gamma^d r_{\max}}{1-\gamma}.$$

U^{random} and U_d^{random} are the random policy's expected returns without and with the depth limit. Increasing d reduces this truncation error; increasing N reduces the sampling uncertainty.

Random Rollout: Cost and Accuracy

Random rollout takes $\mathcal{O}(N \cdot d)$ simulated steps, assuming that choosing and simulating an action takes constant time.

- Apart from storing one estimate per state, the cost does not depend on $|\mathcal{S}|$, which makes rollout attractive for large or continuous state spaces.
- For independent rollouts from s, the variance of the sample mean falls as $\mathcal{O}(1/n(s))$, so the standard error falls as $\mathcal{O}(1/\sqrt{n(s)})$: accuracy improves slowly with more samples.

Even so, a few rollouts can give useful estimates, and the method scales to high-dimensional problems where model-based approaches become intractable.

Example: Value Iteration and Random Rollout

In this example, we return to the slippery 3×3 grid world:

1. Build the MDP and check it against the Bellman backup we worked by hand.
2. Solve it with value iteration, and extract the optimal policy.
3. Estimate each square's value with random rollouts, and compare the estimates with the exact value of the random policy and with U^*.

The rollouts match the random policy, far below the optimal values, and their standard error shrinks only as $1/\sqrt{n}$.

Example: Value iteration and random rollout in the slippery grid world

Lab: Value Iteration in a Grid World

In the lab, a robot must reach a charging station while avoiding lava pits:

- we build the MDP, its reward matrix and transition array, for the robot's grid world,
- we write our own value iteration with a convergence check and compare it with the course package's version, which runs a fixed number of sweeps,
- we extract the optimal policy, follow the robot's path, and see how the discount factor changes its behavior.

Lab: Value iteration in a grid world

Summary

We added actions and rewards to Markov chains, which lets an agent make optimal sequential decisions.

Key takeaways

- **MDPs.** Actions shape where the process goes, rewards score each choice, and a policy assigns an action to every state.
- **Value iteration.** Repeated Bellman backups give the optimal values; the action that achieves each maximum is the optimal policy.
- **Random rollout.** Averaging simulated random trajectories estimates how well random actions do, without the transition probabilities.

Value iteration needs the transition model; random rollout needs only a way to simulate the environment.